

Haoquan Wang

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EDUCATION

University of Utah

Bachelor of Science in Computer Science

Minor: Mathematics

GPA: 3.857

Cumulative GPA Units: 49.000

Relevant Coursework:

Linear Algebra (A) | Discrete Mathematics (A) | Data Structures (A-) | Calculus I & II (A)

Academic Awards:

Dean's List | Fall 2025 – Present

Utah Global Scholarship | Spring 2025 - Present

Salt Lake City, Utah

Jan 2025 - Expected May 2028

WORK

Wuhan Changhang Keda Engineering

Office Staff (40 hours per week)

Wuhan, Hubei, China

August 2022 - October 2023

- Engineered automated data-validation pipelines to audit complex hydrological time-series datasets, identifying compliance anomalies and reducing structural risks.
- Refined statistical cross-referencing algorithms for multi-dimensional geological data, enhancing data integrity and raising joint-distribution accuracy by 3%.
- Evaluated engineering project performance by synthesizing multi-dimensional data trends, providing data-driven insights that mitigated compliance risks by 7%.

PROJECTS

Research Paper: High-Order Temporal Graph Learning for Network Inference | Python, PyTorch

September 2025 - present

- Co-authored working paper proposing a novel high-order temporal graph neural network (HTGN) to infer malicious anomalies in evolving enterprise networks.
- Formulated a permutation-invariant aggregation framework that captures multi-entity, event-level interactions jointly, eliminating structural ambiguity from traditional pairwise decomposition.
- Designed a smooth attention normalization mechanism leveraging Softplus mapping (MultiMax), successfully mitigating SoftMax's concentration-diversity trade-off and softly suppressing low-relevance temporal noise.
- Conducted extensive inductive evaluation on the LANL dataset using an NVIDIA RTX 4090 GPU, consistently outperforming strong temporal graph baselines (Jbail) across AUC, AP, precision, and recall.

Explainable AI (XAI) & Model Robustness Benchmark | Python, PyTorch SHAP LIME

June 2026 - present

- Integrating foundational feature-attribution tools (e.g., SHAP, LIME) to extract and visualize feature importance, enhancing the interpretability of black-box model decisions

Deep Learning: Image Classification with Data Augmentation | Python, PyTorch, CUDA

August 2025 - October 2025

- Implemented a comprehensive CIFAR-10 image classification experiment comparing baseline and augmented training approaches, achieving 4% test accuracy improvement (87% → 91%) while effectively preventing overfitting.
- Engineered GPU-accelerated training pipeline using PyTorch with CUDA 11.8, optimizing batch processing and mixed-precision training to achieve 10-20x speedup over CPU implementation.
- Applied multiple data augmentation techniques (RandomHorizontalFlip, RandomCrop, ColorJitter, RandomRotation) combined with Dropout and BatchNormalization, reducing overfitting gap from 13% to near-zero and significantly enhancing model generalization on unseen data.

Autonomous Path-Finding Drone | C++, Arm Mbed OS

July 2023 - March 2024

- Engineered a navigation drone using C++ on the Arm Mbed OS platform, implementing real-time sensor data processing for autonomous flight.
- Optimized control algorithms to bridge the gap between high-level path planning and low-level motor execution, significantly improving flight stability.

Data Structures & Algorithms Implementation | Java (GT CS 1332 Curriculum)

May 2025 - September 2025

- Developed 10+ core data structures from scratch, including External Chaining HashMaps, AVLTrees, and MaxHeaps, adhering to GT's strict $O(\log n)$ efficiency standards.
- Implemented complex graph algorithms such as Dijkstra's and Prim's, ensuring robust memory management and algorithmic optimization.

SKILLS

Languages: Python, Java, C/C++

Tools: Pandas, PyTorch, TorchVision, NumPy, Mbed OS, Git, JUnit, SHAP